

Date: Jan 25, 2021

Experiment: 3

Title: Simple distillation

Objective

To purify a contaminated liquid sample by simple distillation

Theory

Simple distillation is a procedure by which two liquids with different boiling points can be separated. Simple distillation (the procedure outlined below) can be used effectively to separate liquids that have at least fifty degrees difference in their boiling points. As the liquid being distilled is heated, the vapours that form will be richest in the component of the mixture that boils at the lowest temperature. Purified compounds will boil, and thus turn into vapours, over a relatively small temperature range (2 or 3°C); by carefully watching the temperature in the distillation flask, it is possible to affect a reasonably good separation.

Apparatus

- Condenser
- Thermometer
- Measuring cylinder (both as a liquid measuring device and as a receiving flask)
- Iron ring
- water bath
- Stand
- Distillation flask
- Boss head
- Beamless funnel
- Bunsen burner
- Clamp

Chemical used

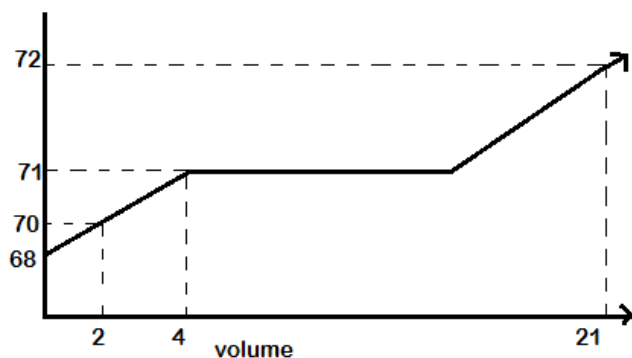
- Ethyl alcohol
- Boiling chips
- water

Procedure

The simple distillation apparatus was assembled by following the instruction of our instructor. 25ml of impure ethyl alcohol was measured and added into the distillation flask. 3 pieces of boiling chips were added into the distillation flask. The distillation flask containing the impure ethyl alcohol was placed in a water bath. The distillation flask was heated indirectly by heating the water bath. The ethyl alcohol was boiling, and at 68°C the first droplet was observed. 2ml of pure ethyl alcohol was collected at 70°C.

Result and discussion

Volume (ml)	Temperature (°C)
1 st droplet observed	68
2	70
4	71
6	71
8	71
„	71
„	71
21	72



The data collected throughout the experiment was used to construct a distillation curve for simple distillation; the data shows that the distillation yield for each distillation is increasing evenly over time. However this experiment graph shows unpredictable pattern for the amount of distillation being collected. In the data collected throughout the experiment revealed that the most effective method of distillation method of distillation for this experiment was simple distillation. This is could be due to the fact that simple distillation is more effective when the liquid in a mixture has a larger difference in boiling points. The experiment resulted in simple distillation at 72°C.

Quiz questions

1. Why does not all the liquid vaporize at once when the boiling point is reached?

- Because not all molecules go from the liquid to gaseous state immediately. The vapour will form nearest the flame because the heat energy will speed up the kinetic energy (movement) of those molecules first. Subsequently the vapour rises to the top and new

liquid molecules get closer to the flame. As a side, you may also see bubbles forming on the sides and other parts of the flask. Remember, liquid molecules that are just ready to vaporize are still moving and begin to show in other places than just over the flame. Also, heat will conduct through the glass flask and cause other molecules to vaporize in other places.

2. An organic liquid begins to decompose at 80°C. Its vapour pressure at that temperature is 36mm. How would you distil the liquid?

- The organic liquid could be distilled by steam distillation, because this distillation is used for high boiling point substances that decompose before the boiling point is reached. In this case, instead of using a vacuum, the liquid in question is mixed with another, immiscible liquid. The presence of the second liquid causes both to boil at a temperature lower than the regular boiling points of either liquid.

3. Why is cold water circulated through a condenser from the bottom upward rather than from the top downward?

- Water is circulated from the bottom upward in order to fill the entire condenser. If water enters from the top it drains from the bottom and never fills it up. A full condenser is necessary to maximize the heat transfer for condensation. If the water enters through the top and the flow rate is insufficient, the water will merely trickle down one side of the condenser. This would mean only a small section of the condenser would be adequately cooled enough to provide condensing action. By allowing water to enter from the lower hole, the entire volume of the chamber would have to be filled before the water left, provide far more efficient cooling.

4. If the liquid to be distilled had a boiling point higher than 90°C, what kind of bath would you use?

- Water baths, heated on a hotplate, are most commonly used to heat solutions. They may also be used to heat to lower temperatures, although it can be difficult to maintain a constant temperature.
- Oil baths are much like water baths, but use silicone or mineral oils in order to enable temperatures hotter than the boiling point of water (> 100°C). So it is preferable to use an oil bath since the liquid has a high boiling point.

5. Why is it that when a pure liquid is boiled its temperature remains constant?

- The temperature remains constant during boiling of water even though heat is supplied constantly because all the heat energy provided is used up in changing the state of water from liquid to gaseous water vapour. Therefore all the heat energy provided to the liquid increases the kinetic energy of the particles and temperature doesn't increase. So supplied heat energy is used to increase kinetic energy of molecules. The extra heat supplied is supplied to the boiling water is used in the vaporization of a liquid, which is known as latent heat of vaporization. That's why the temperature of a substance remains constant during its melting point.



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